Bendicht Grossniklaus

Embedded Software Engineer

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PROFESSIONAL PROFILE

Embedded Software Engineer with hands-on experience in electronics, mechatronics and industrial product development. Specialized in embedded C development for microcontroller-based systems, with a focus on real-time motor control, LoRaWAN communication and ultra-low-power applications. Experienced in customer-specific engineering projects, quickly adapting to different development environments, workflows and technical requirements. Bringing experience from diverse customer projects, covering requirements definition, firmware implementation, system integration, troubleshooting, testing and validation.

WORK EXPERIENCE

2021 - Present **Electronic Development Engineer**, *Naxys GmbH*, Prés de l'Envers 2, 2607 Cortébert

Consultant at Sontex SA, contributing to the development and integration of ultra-low-power LoRaWAN communication capabilities into industrial products. Product data is encoded in the M-Bus format and transmitted wirelessly via LoRaWAN.

- **Modular LoRaWAN platform development (2021):** Developed a reusable firmware platform in C based on the STM32WLE5CBU6 microcontroller. The dedicated communication controller is integrated into the product PCB and exchanges data with the main product microcontroller, enabling LoRaWAN connectivity across different products while keeping modifications to existing product firmware to a minimum. To achieve ultra-low-power operation, RTC events are managed by the product's main firmware, allowing the LoRaWAN controller to enter a low-power mode with a current consumption of approximately 0.7 μA until woken up via GPIO.
- **Plug-on module development (2023):** Developed a plug-on LoRaWAN communication module for an existing Sontex product by creating a dedicated firmware variant of the previously developed platform based on the STM32WLE5CBU6. Since the module manages its own RTC events, it operates in a different low-power configuration with a baseline current consumption of approximately 4 μA.
- **Product range extension:** Extended the Sontex product range with LoRaWAN connectivity by configuring the reusable platform and adapting the firmware in C for products based on STM8L052R8 **(2021-2022)**, STM32L053R8T **(2024-2025)** and MSP430FR5986IPMR **(2026)**. Designed and implemented solutions for product-specific requirements while maintaining the low-power constraints of each device.
- **Specification, testing and validation:** Used Jira and Confluence for technical documentation, specification writing, bug reporting and test case definition. Prepared and executed firmware validation test plans for different products prior to production release.

2020 - 2021 **Control Systems Engineer**, *Naxys GmbH*, Prés de l'Envers 2, 2607 Cortébert

Modernized the control system of a customized heating installation serving a multi-family residential building.

- Evaluated and selected a suitable PLC based on the interfaces and compatibility requirements of the existing hydraulic system and field devices, including motors, valves and sensors.
- Reimplemented the control software using CODESYS.
- Developed control algorithms for solar thermal heating, wood-fired heating and heat distribution across four residential zones.
- Commissioned and integrated the new control system with the existing heating infrastructure.
- Performed functional testing and validation of the complete system.

2019 - 2020 **Electronic Development Engineer**, *Société industrielle de Sonceboz SA*, Rue Rosselet-Challandes 5, 2605 Sonceboz-Sombeval

Developed the software and electronics for an actuator driving a Continuously Variable Transmission (CVT) for agricultural engines, as part of a new actuator family for hydraulic valves. Collaborated closely with the customer and project manager to bring the actuator from prototype to pre-series stage.

2016 - 2019 **Electronic Development Engineer**, *Sonceboz Microtechnique Boncourt SA*, Rue des Boulaies 2, CH-2926 Boncourt

Contributed to the industrialization of CPM products in the Truck and Offroad division, supporting product development, customer projects and production.

- Embedded software development in MISRA-compliant C on the SPC560P44L3, including position control, analog speed throttle integration and issue resolution.
- Introduction of new PCB revisions and coordination of prototype assembly.
- Technical support for customers worldwide.
- Investigation and resolution of product and production issues in collaboration with manufacturing teams.

2015 - 2016 **Student Internship**, *Société industrielle de Sonceboz SA*, Rue Rosselet-Challandes 5, 2605 Sonceboz-Sombeval

Investigated the feasibility of estimating BLDC motor coil temperatures through resistance measurements derived from applied voltage and current. Implemented the software and conducted measurement campaigns to evaluate the method, demonstrating its potential for approximate temperature estimation. Documented the results in a laboratory journal as a basis for potential patenting.

2014 **Civilian Service**, *Ecological Engineering Laboratory (ECOL)*, EPFL, CH-1015 Lausanne

Worked in a team to improve the robustness and add new functionalities to an autonomous motorized catamaran based on a NI computer programmed in LabVIEW. Also participated in data collection campaigns on Lake Geneva, supporting doctoral research projects.

2012 - 2013 **Freelance Translator**, *SABAG LUZERN AG*, CH-6023 Rothenburg

Translated a tutorial from German into French to introduce employees to a new logistics software program.

2008 - 2009 **Civilian Service**, *CH-4000 Basel Hörnli, CH-2608 Prés-de-Cortébert, CH-2503 Biel/Bienne*

Different assignments with a total duration of seven months in landscape gardening, organic farming and traditional dry-stone wall construction.

EDUCATION

2014 - 2016 **Master**, *Ecole Polytechnique Fédérale de Lausanne (EPFL)*, CH-1015 Lausanne

Micro-engineering, specialization in Robotics and Autonomous Systems (RSA)

2016 **Master Thesis: Design of an external BLDC motor with integrated electronics for 48V on-board power supply**, *Sonceboz SA - Integrated Actuators Laboratory (LAI)*, CPM Compact Power Motion GmbH, Feringastraße 11A, 85774 Unterföhring, Germany

Conducted a Master Thesis in collaboration between the SONCEBOZ GROUP and LAI-EPFL at Compact Power Motion GmbH (CPM) in Munich, focusing on the design of an external-rotor BLDC motor with twice the power and torque of existing CPM products. Designed and optimized motor geometry using Flux2D FEM simulations, performed thermal analysis in SolidWorks and developed a Software-in-the-Loop model in MATLAB/Simulink. Achieved a design meeting the defined requirements for torque, power and motor control.

2015 **Semester Project II: Automation of a laboratory coil winding machine**, *Integrated Actuators Laboratory (LAI)*, Rue de la Maladière 71 B, CH-2002 Neuchâtel

Automated a laboratory coil winding machine using an Arduino and a linear stepper motor. Implemented synchronized motor control using encoder feedback and a simple LCD-based user interface.

2014-2015 **Semester Project I: Development of a soft jellyfish type underwater robot**, *Laboratory of Intelligent Systems (LIS)*, EPFL, CH-1015 Lausanne
Designed and tested a jellyfish-inspired underwater robot using Dielectric Elastomer Actuators (DEAs). Developed MATLAB models and evaluated the prototype's propulsion, deformation and power consumption.

2009 - 2014 **Bachelor**, *Ecole Polytechnique Fédérale de Lausanne (EPFL)*, CH-1015 Lausanne
Micro-engineering

2008 **High school diploma**, *Gymnasium Alpenstrasse*, CH-2502 Biel-Bienne
Physics and applied mathematics, bilingual (French-German)

TECHNICAL SKILLS

Programming Languages	Core competency: **C** (extensive hands-on experience, including MISRA-C) Additional experience: Structured Text (IEC 61131-3), Vector CAPL, MATLAB/Octave, C++ (Qt Framework) Basic knowledge: Python, Rust
Embedded Concepts	**Bare-metal programming**, real-time programming, time-triggered scheduling, low-power management, interrupt-driven programming, event-driven state machines, error handling.
Microcontrollers	MSP430FR5986IPMR, STM32WLE5CBU6, STM32L053R8T, STM8L052R8, SPC560P44L3.
Peripherals	GPIO, ADC/DAC, PWM, DMA, interrupts, timers, RTC, watchdogs, low-power modes.
Communication	**LoRaWAN**, M-Bus, UART, SPI, I²C, CAN, LIN.
Debugging & Measurement	GDB-based debugging, JTAG/SWD, ST-Link, J-Link, oscilloscope-based debugging and protocol analysis (PicoScope), current consumption measurements (Keysight power analyzer, Nordic Power Profiler II).
Development Tools	Eclipse-based IDEs (STM32CubeIDE, CCS), Git (TortoiseGit), Atlassian Jira/Confluence, SpiraTeam, Vector CANalyzer, Altium Designer, SolidWorks, MATLAB/Simulink, Flux2D/3D, LabVIEW, LaTeX, MS Office.
Hardware-related Skills	Basic electronic design, PCB prototype debugging, soldering, building small test and measurement setups, drilling and milling.

LANGUAGES

German	Bilingual (Swiss-German native language)
French	Bilingual (Obligatory school)
English	Good knowledge (Level B2)
Spanish	Good oral knowledge

HOBBIES AND INTERESTS

Sports	Capoeira, kitesurfing, biking, snowboarding, cross-country skiing, kayaking Swiss junior national team kayak slalom (2003-2007), J+S guide in kayak (2007)
Handicraft work	Transformation of a van into a camper, building and operating an aquaponic system, permaculture and traditional gardening